PARTNER BRIEF · DRAFT FOR DISCUSSION

Calibrating AFM with DNA origami: an open method, with its uncertainty

dobladillo project · 29 September 2026 · open source (EUPL-1.2), self-hosted tools only



0 nm height, as an AFM draws it 20 nm

An AFM image of anything smaller than 100 nm has no good ruler to be checked against: commercial pitch gratings stop at about 70 nm. DNA origami puts features at designed positions a few nanometres apart, and it can be imaged in the same buffer as the sample. The objects already exist, some of them for sale. What does not exist is an open, validated way to turn an image of them into calibration values *with an uncertainty*. That is what we propose to build, and where we would like partners.

ALREADY EXISTS

Origami AFM samples are sold: a ruler rectangle and a tip-reconstruction staircase (GATTAquant). One metrology institute has published an AFM study (VTT MIKES, 2017). None carries certified values.

OPEN

Open software that reports scale, skew, tip, resolution and drift with uncertainties; AFM traceability of origami dimensions; a reference sample built for high-speed AFM in liquid.

WHAT WE ASK

Instrument time and images, one traceable reference measurement, and eyes on the method. Details in section 7.

1 The calibration gap below 100 nm

An AFM's lateral scale is set against a pitch grating. The smallest commercial ones are around 70–150 nm pitch, and a field about 100 nm wide cannot be calibrated with a grating much coarser than that [Dixson 2010]. Atomic lattices (mica, graphite) work at a few nanometres, but small ambient scans are dominated by drift [Kim 2022]. Between those two lies the scale at which proteins and DNA nanostructures are imaged, and most of it is calibrated by extrapolation.

The tip is the other unknown. It broadens every feature by several nanometres, it wears during a session, and it is usually characterised on a separate hard sample, not in the image that is being measured. High-speed AFM has no reference sample at all; laboratories improvise with collagen, streptavidin crystals or graphite [Marchesi 2021; Umeda 2023].

DNA origami is a natural candidate for both. Its features sit at positions set by base-pair counts, it deposits alongside the specimen, and it images in liquid. Its weakness is equally clear: its dimensions have been reported to vary by 3–10 %, "too high for a

proper metrological calibration standard", most likely because of deposition and drying rather than the structures themselves [Graugnard 2017], and its apparent height depends on imaging force and buffer. It can be a *transfer* standard, calibrated once against a traceable reference; it is not a primary one, and it is not a height standard.

2 What others have done

WORK	WHAT IT PROVIDES	STATUS
GATTA-AFM Nanoruler (GATTAquant)	Origami rectangle, about 90×70 nm, with a line ladder of about 6 nm spacing; in solution, ≥10 surfaces; €350.	PRODUCT not certified
GATTA-AFM 3D (GATTAquant); Kolbeck et al., *Nano Lett.* 2023	Four-step staircase, about 2.5/5/12.5/20 nm, for blind tip reconstruction; about ten copies give a usable tip; €450.	PRODUCT not certified
Korpelainen et al. (VTT MIKES), *Meas. Sci. Technol.* 2017	Seeman tile as an AFM xyz standard: height 2.0 ± 0.2 nm, width 88 ± 3 nm; dry samples stable ≥12 months.	PUBLISHED
Raab et al. (PTB), *Sci. Rep.* 2018	Origami distances made SI-traceable, by DNA-PAINT fluorescence, not AFM.	PUBLISHED
Kielar et al., *Angew. Chem.* 2020	Origami used to measure how invasive the high-speed AFM tip is; streptavidin markers are stripped by the tip over minutes.	PUBLISHED
EMRP SIB61 (PTB, MIKES and others)	Named DNA origami as a candidate lateral and vertical standard; no product followed.	PROJECT
Gwyddion, TopoStats, NanoLocz	Open AFM analysis: Gwyddion does blind tip estimation and lattice-based distortion correction; none ships origami reference geometry or reports calibration with uncertainty.	SOFTWARE

> From a literature and product review made on 29 September 2026. "Not found" below means our searches turned nothing up, not that nothing exists; we would welcome corrections.

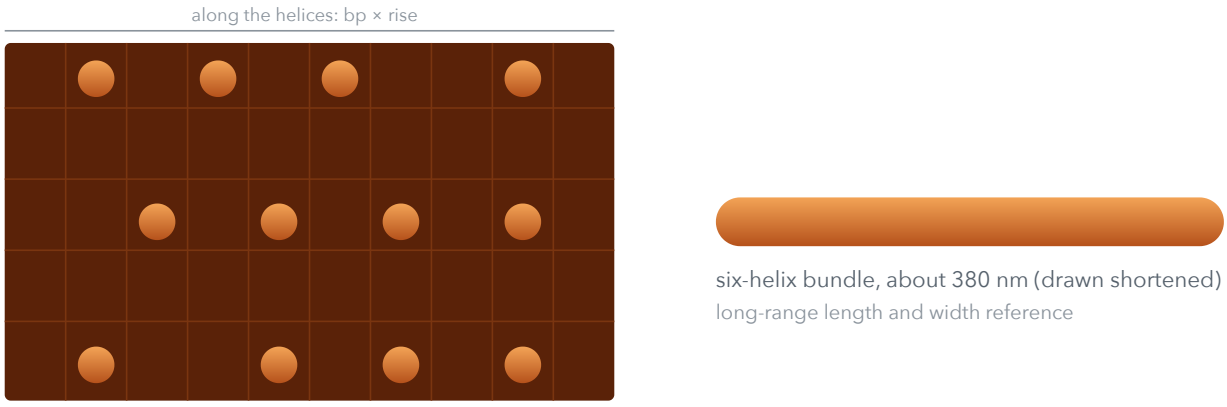
3 What is open

GAP	WHY IT MATTERS
OPEN Calibration software with uncertainty	Existing samples come with nominal dimensions and no analysis. A user cannot say "my x scale is 1.03 ± 0.01" from them without writing the method themselves.
OPEN AFM traceability of origami dimensions	The only AFM uncertainty budget is VTT 2017; traceability has been done for fluorescence only. No certified reference material or interlaboratory comparison exists.
OPEN A reference built for high-speed AFM in liquid	Protein markers do not survive the tip. An all-DNA reference, readable at any landing orientation, has not been offered.
MINOR Graded resolution ladder	Only a fixed 6 nm ladder exists; graded spacings would read the resolution directly.

4 The proposal: a method, not another sample

The kit has four parts. The software comes first because it fills the clearest gap, needs no laboratory to start, and works with samples that can already be bought.

PART	WHAT IT IS
Software	`dobladillo calibrate` (working, validated on synthetic scans): reads a scan of any reference whose geometry is declared (ours, GATTAquant's, Kolbeck's) and reports lateral scale, skew, tip shape, resolution and drift, each with an uncertainty. Open source, runs locally.
Reference samples	A *calibration tile*: a DNA-origami rectangle carrying a sparse, chiral pattern of hairpin marks whose centres are the ruler, all DNA, readable whichever way it lands. A *six-helix bundle* of about 380 nm as a long-range length and width reference.
Protocol	Folding, purification, deposition and imaging specified tightly enough that the 3–10 % preparation spread is measured and bounded, not assumed away.
Certificate method	How a batch's dimensions are tied to a traceable reference, with a GUM-style uncertainty budget. Until then every output carries `traceable: false`.



calibration tile: 12 marked cells, 48 hairpin staples · the actual design

The two references. Each mark on the tile is a cell of hairpins; its centre, not its edge, is what is measured, because tip broadening widens a mark symmetrically and does not move its centre. No two marked cells touch, so each images as its own spot, and the pattern is chiral, so every landing orientation is identified. The mark centres span 75.5 × 38.8 nm.

Why the scale can be separated from the sample

Origami tiles land at random angles. The scanner's distortion is fixed in the image frame, while the tile's own anisotropy (its helix spacing, which depends on buffer) turns with each tile. Fitting many tiles at once therefore separates the two: the x and y scale and the skew of the scanner, and the true helix spacing of this preparation. The software refuses the fit when the tiles' angles are not varied enough to tell them apart.

5 What the software reports

QUANTITY	FROM	METHOD
x and y scale, skew	Mark centres on many tiles at varied angles	Joint least-squares fit of scanner and tile; bootstrap over tiles; combined with the reference's own uncertainty
Tip shape, apex radius	Blind: any image with features sharper than the tip. Known geometry: a declared reference such as the staircase	Villarrubia (1997), implemented from the paper; a blind estimate is an outer bound, never sharper than the true tip, and a scan that cannot constrain it is flagged
Lateral resolution	The reconstructed tip and the image noise	Smallest separation at which two marks show a dip deeper than the noise; checked directly where marks are paired
Drift	A series of frames (high-speed AFM or repeated scans)	Sub-pixel cross-correlation between frames; linear drift velocity with its standard error

Validation so far: synthetic scans with a known answer

TEST	PUT IN	RECOVERED
Scanner and tile, 40 tiles	scale x 1.05, y 0.97, skew 1.5°, tile 1.08	1.0499, 0.9700, 1.501°, 1.0800 (statistical u ≤ 0.0003; 0.013°)
Tip insensitivity	the same tiles, 5 vs 15 nm tip	mark centres move 0.11 nm rms; outlines widen 4.9 nm
Honest intervals	30 independent sessions	truth inside the k = 2 interval 27–28 times; the bootstrap reads about 15 % low, and the report says so
Tip, blind	paraboloid tips of 5, 10, 20 nm, sharp spikes	5.5, 10.5, 20.5 nm ± 0.55 without noise; with noise an upper bound, 11–53 % high
Resolution	10 nm tip, 0.1 nm noise, 2 nm features	5.1 nm, within 1 % of the closed-form sphere–cylinder result
Drift	0.30 and −0.20 nm/s over 20 frames	0.3001 ± 0.0002 and −0.1998 ± 0.0001 nm/s
Refusals	tiles all at one angle; too few tiles; plain rectangles; a featureless scan	each refused or flagged with the reason, never a confident number

Two findings from this stage shape the plan. The tile's size across the helices cannot be taken from the design, because published helix spacings range over 2.25–2.6 nm; the fit therefore measures it, and the x/y ratio and skew do not depend on it. And a staircase of flat blocks gives *blind* reconstruction nothing to work with, since a flat top with sloped sides is consistent with any blunter tip; it constrains the tip through its declared geometry instead. The published staircase work reports blind reconstruction from real images, so experiment K4 tests this on real data.

Every value is reported with its standard uncertainty and a k = 2 interval. Scans are read in their native formats (Bruker, JPK, Asylum, Gwyddion) through the open AFMReader library. No real image has been analysed yet: that is K3 onwards. Not built yet: a report that reads a whole experiment folder at once, high-speed AFM movie formats (frames are exported one by one), and grating pitch analysis (Gwyddion does it).

6 Experiments

Each experiment has a written protocol, a named readout computed by the software, and a pass criterion that the project lead writes before the data exist. Desk work needs no laboratory.

	EXPERIMENT	WHAT IT DECIDES	NEEDS	WHERE
K1	Software validation	Whether the software recovers a known distortion, tip and drift from synthetic scans and public data, with honest intervals.	—	DESK
K2	Fold, purify, yield of the tile	Whether the tile folds cleanly enough to be a reference; isothermal folding as a cheaper route.	E1, E8, E11	LAB
K3	First lateral calibration	Mark detection rate in air and liquid; agreement with a traceable grating at larger scan size.	K2	AFM
K4	Tip reconstruction	Whether our tile and bundle give the same tip as the published staircase with the same tip; tip wear over a session.	K2	AFM
K5	Preparation sensitivity	How much magnesium, nickel, drying and deposition time move the dimensions: this becomes the uncertainty.	K3	AFM
K6	High-speed AFM in liquid	Scale and drift in buffer; whether all-DNA marks outlast protein markers under the tip.	K2, K3	HS-AFM
K7	Kit format and shelf life	Frozen aliquots or dried on mica; freeze–thaw tolerance.	K2	LAB
K8	Certification	A batch's dimensions tied to a traceable reference with a full uncertainty budget.	K3, K5	PARTNER

7 What we ask of partners

- **Instrument time and images.** Scans of the tile and the bundle in air and in liquid, with the instrument's own calibration record, so the software can be tested against something real.
- **A high-speed AFM session** for K6: frames of the tile in buffer over tens of minutes.
- **One traceable measurement** for K8, from a laboratory with a metrological AFM or a traceable grating at the relevant scale.
- **Review of the method**: the uncertainty budget, the choice of criteria, what an AFM user would need the report to say.

In return: the designs, protocols and software are open (EUPL-1.2), results are shared with contributors, and authorship follows contribution. Nothing depends on an external service; every tool runs locally.

8 Limits we state up front

- **Not a height standard.** DNA's apparent height depends on force, mode and buffer; the kit can report it, not calibrate against it.

- **Not traceable until certified.** Gratings and metrological AFMs reach parts in 10^4; uncertified origami is good to a few per cent.
- **Preparation matters.** The 3–10 % spread is the main risk. The protocol and K5 exist to measure and bound it.
- **Not a competitor to existing samples.** The software reads commercial samples too; its value is the method and the uncertainty.

9 References

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